

Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

General Remarks

All members participated in the discussion of the empirical approach, interpretation of results, code review, and final editing. All parts were jointly reviewed by the group. To make the individual contributions transparent, the primary responsibility for the different parts is distributed as follows: Exam number 45: ex. 1 & 4. Exam number 46: ex. 2 & 5. Exam number 47: ex. 3.

The use of generative AI tools is documented in the submitted GAI declaration.

Exercise 1: Choice of realized variance estimator

We first load the minute-level closing price data during regular trading hours and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp. We place AAPL prices on a fixed one-minute calendar-time grid from 09:30 to 15:59, corresponding to 390 observations per normal trading day. Missing prices are handled by previous-tick interpolation within each trading day only: if no price is observed in a given minute, the most recent price from the same day is carried forward. Before interpolation, AAPL's missing share on this grid is 0.40%. This creates a complete intraday grid, but may also introduce stale prices in minutes without new trades. We therefore do not carry prices across trading days or backfill before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, prices are sampled at pre-specified timestamps relative to the market open, e.g. 09:30, 09:40 for 10-minute sampling. A bucket-based approach would instead typically use the last observed price within each interval, e.g. 09:39, 09:49 for 10-minute buckets. We use fixed-grid sampling to make the timing convention explicit and to ensure that intraday returns are aligned across stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where $r_{d,j}^{(\Delta)}$ is the intraday log return over sampling interval.

As shown in Figure 1, the average realized variance is highest at the shortest sampling intervals and falls substantially when moving from 1-minute to 5-minute sampling. This is consistent with market microstructure noise at very high frequencies. In the lecture slides, the observed log price is written as the latent efficient log price plus a noise component, $Y_{i\Delta_n} = X_{i\Delta_n} + U_{i\Delta_n}$, and the noise-driven fraction of return variance increases as Δ_n becomes small. Sampling less frequently therefore reduces the influence of bid-ask bounce, price discreteness and stale prices. However, very long sampling intervals discard useful intraday information.

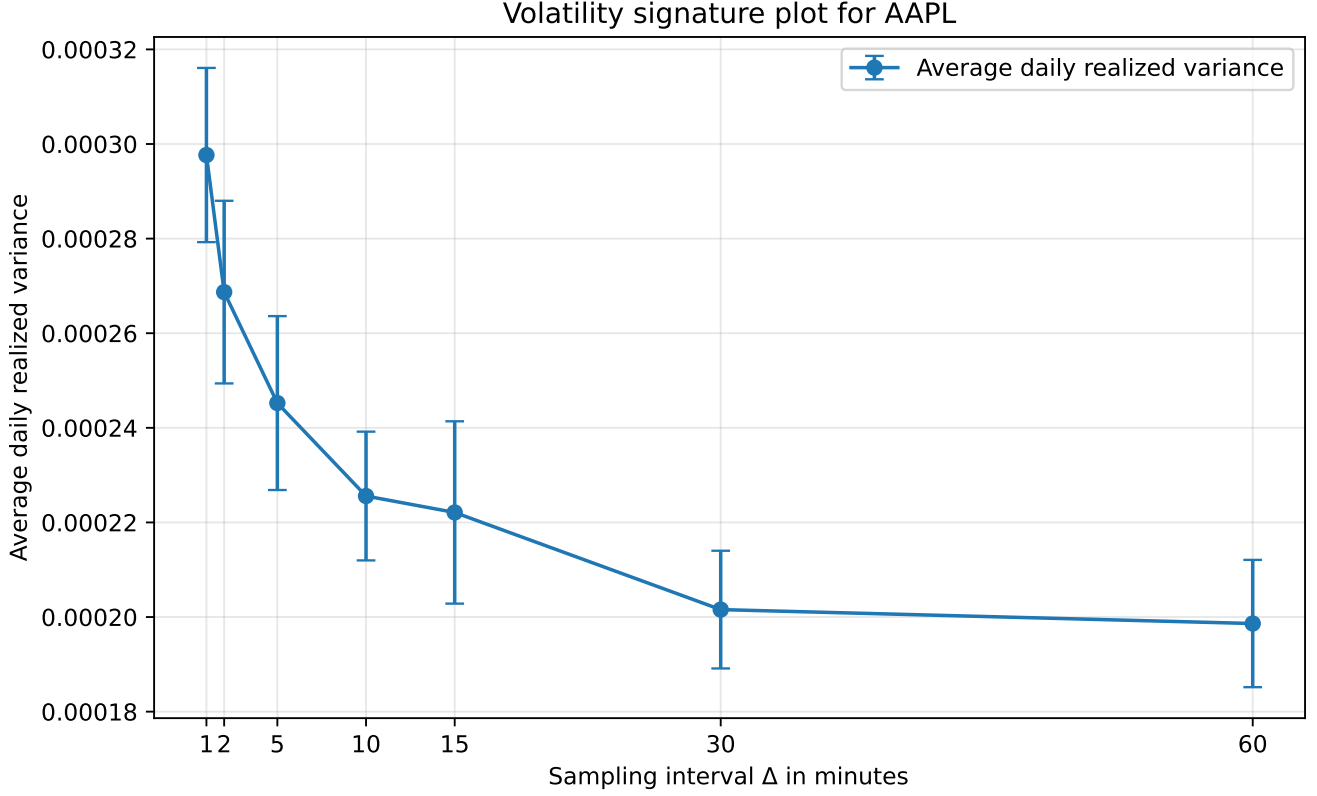


Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

We choose the standard realized variance estimator based on 5-minute log returns for the remainder of the analysis:

$$RV_d^{(5)} = \sum_j \left(r_{d,j}^{(5)} \right)^2.$$

This choice is also consistent with the empirical realized-volatility literature, where 5-minute returns are commonly used as a practical compromise between exploiting high-frequency information and limiting market microstructure noise (Andersen and Bollerslev, 1998). At the same time, 5-minute sampling leaves approximately $390/5 - 1 = 77$ intraday returns per trading day. Since the subsequent realized covariance matrix is 30-dimensional, this helps avoid the rank problem emphasized in the lecture slides: when the return matrix has $T < N$, the sample covariance matrix is singular and cannot be inverted. Thus, 5-minute sampling is preferred to lower-frequency choices such as 15, 30 or 60 minutes, which leave fewer intraday returns for estimating the 30×30 covariance matrix.

Exercise 2a: Daily realized covariance matrices

We estimate the daily realized covariance matrix for the 30 DJIA stocks using the standard realized covariance estimator:

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j},$$

where $r_{d,j}$ denotes the 30×1 vector of synchronized intraday log returns at interval j on trading day d . This estimator is the multivariate analogue of the realized variance estimator used in Exercise 1.

Following Exercise 1, covariance is estimated using synchronized 5-minute returns. The same sampling frequency is used throughout the remainder of the analysis, ensuring consistency with the realized variance estimator selected based on the volatility signature plot.

To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day and missing observations are handled using previous-tick interpolation (forward filling). Returns are then computed on a common 5-minute grid, ensuring that the covariance matrix is based on synchronized return vectors. This synchronization is important because non-synchronous trading is a key challenge in multivariate high-frequency covariance estimation ([Barndorff-Nielsen et al., 2011](#)).

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i , the diagonal element is:

$$RC_{d,ii} = \sum_{j=1}^{M_d} r_{d,j,i}^2.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

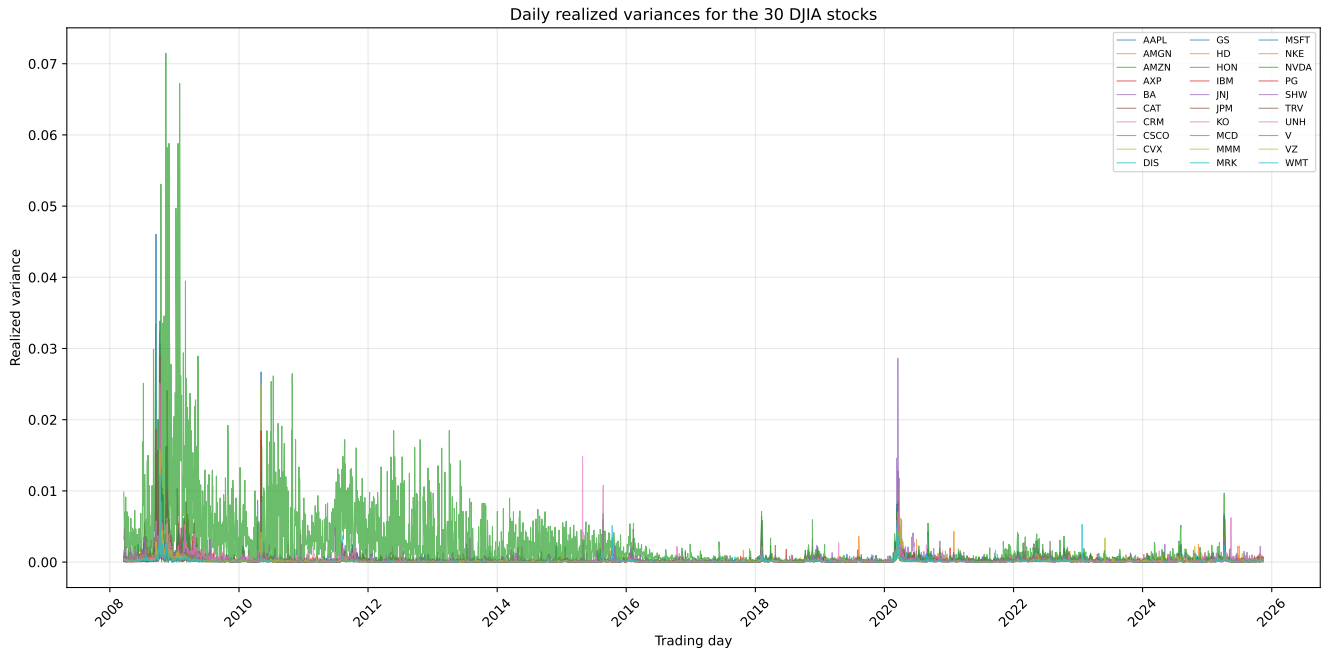


Figure 2: Daily realized variances for the 30 DJIA stocks.

As shown in Figure 2, the realized variance series vary substantially over time and exhibit clear volatility clustering. Periods with elevated realized variance tend to be followed by further volatile periods, while calmer periods are associated with persistently lower realized variance. The largest individual realized variance spike occurs for NVDA on 2008-11-13, with a realized variance of 0.0715.

The figure also shows that volatility spikes often occur across several stocks at the same time, suggesting a common market-wide volatility component. The day with the highest average realized variance across the 30 stocks is 2008-10-10. At the same time, there are clear cross-sectional differences in volatility: NVDA has the highest average realized variance over the sample, while JNJ has the lowest.

By sampling at a one-minute frequency instead, the off-diagonal elements of RC_d would likely be more affected by market microstructure noise. As discussed in the lecture slides and [Barndorff-Nielsen et al. \(2011\)](#), covariance estimates at very high frequencies can suffer from the Epps effect, whereby the measured dependence between assets is biased downward. Although the DJIA data are recorded on a common one-minute grid and therefore largely mitigate the non-synchronous trading problem, one-minute returns remain more sensitive to bid-ask bounce and price discreteness. Consequently, the off-diagonal elements of RC_d may become noisier and potentially lower than those obtained from synchronized five-minute returns. This provides additional support for the five-minute sampling frequency selected in Exercise 1.

Exercise 3: Covariance estimators

For each day t in the second half of the sample, we construct two estimators of today's covariance matrix using only information available up to day $t - 1$.

Exercise 3a: High-frequency random walk estimator

The high-frequency estimator is defined as

$$\hat{\Sigma}_t^{HF} = RC_{t-1}.$$

Thus, yesterday's realized covariance matrix is used as the forecast of today's covariance matrix. This implies a random-walk assumption for covariance dynamics: conditional on information available at day $t - 1$, today's covariance is best approximated by yesterday's realized covariance. The assumption is motivated by the persistence and clustering typically observed in volatility, but the estimator may be noisy because it relies only on the previous day.

Exercise 3b: Ledoit-Wolf shrinkage estimator

For the Ledoit-Wolf estimator, we first construct daily close-to-close log returns for all 30 stocks using the last observed intraday price on each trading day. For each evaluation day t , the covariance matrix is estimated from a rolling window of $W = 252$ daily returns ending on day $t - 1$. This corresponds approximately to one trading year and provides enough observations to estimate a 30-dimensional covariance matrix while still allowing the estimator to adapt to changing market conditions.

The Ledoit-Wolf estimator shrinks the sample covariance matrix toward a structured target, reducing estimation error and improving the stability of the covariance matrix used for portfolio construction (Ledoit and Wolf, 2003, 2004). The resulting matrix is stored as $\hat{\Sigma}_t^{LW}$, and its diagonal elements are used in the comparison in Exercise 3c.

Exercise 3c: Comparing realized and Ledoit-Wolf variances

To compare the smoothing properties of the two estimators, we focus on the diagonal elements of the covariance matrices. For each stock i , we compare the Ledoit-Wolf implied variance, $\hat{\Sigma}_t^{LW}t, ii$, with the realized variance, RCt, ii , from Exercise 2b.

As shown in Figure 3, the realized variances are much more volatile than the Ledoit-Wolf implied variances. The realized variance series contain sharp daily spikes because they are computed directly from intraday returns and react immediately to short-run volatility shocks. In contrast, the Ledoit-Wolf estimates are smoother and move more gradually over time, reflecting the 252-day rolling window and the use of shrinkage. This is consistent with the purpose of Ledoit-Wolf shrinkage, which is to reduce estimation error by pulling noisy covariance estimates toward a more stable target (Ledoit and Wolf, 2003, 2004). Thus, the comparison shows that realized variances react quickly to new information, while Ledoit-Wolf provides a more stable but slower-moving estimate of risk.

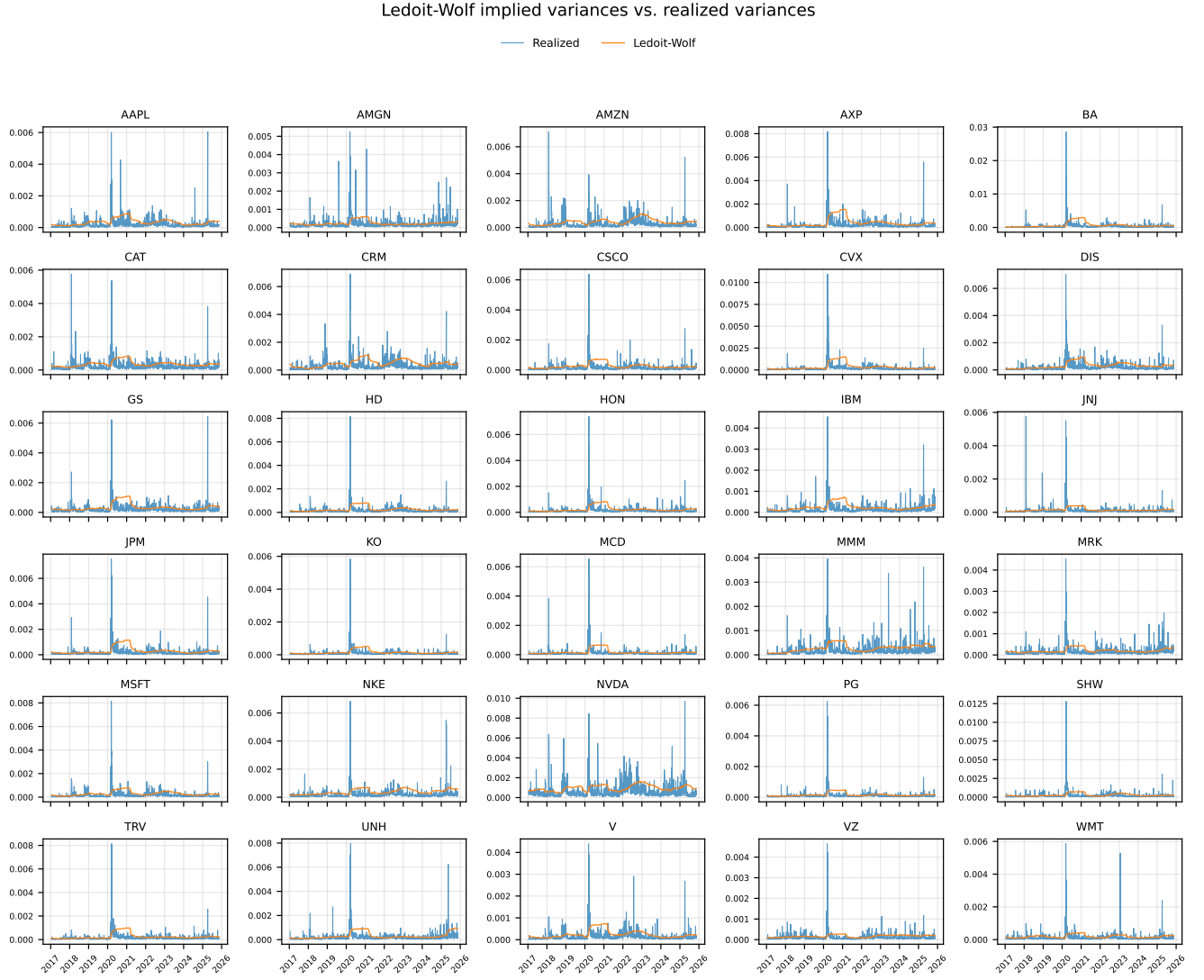


Figure 3: Ledoit-Wolf implied variances and realized variances for the 30 DJIA stocks. The Ledoit-Wolf estimates are based on a 252-day rolling window of daily close-to-close returns, while realized variances are the diagonal elements of the daily realized covariance matrices.

Exercise 4: Minimum-variance portfolios

Using the covariance forecasts from Exercise 3, we compute daily unconstrained minimum-variance portfolios for both the high-frequency random-walk estimator, $\hat{\Sigma}_t^{HF}$, and the Ledoit-Wolf estimator, $\hat{\Sigma}_t^{LW}$. For a given covariance estimate $\hat{\Sigma}_t$, the portfolio solves:

$$\omega_{mvp,t} = \arg \min_{\omega_t} \omega_t' \hat{\Sigma}_t \omega_t \quad \text{s.t.} \quad \iota' \omega_t = 1,$$

which gives the closed-form solution

$$\frac{\hat{\Sigma}_t^{-1} \iota}{\iota' \hat{\Sigma}_t^{-1} \iota},$$

where ι is a vector of ones. Comparing the resulting weights shows how the two covariance estimators translate into portfolio allocations and rebalancing needs.

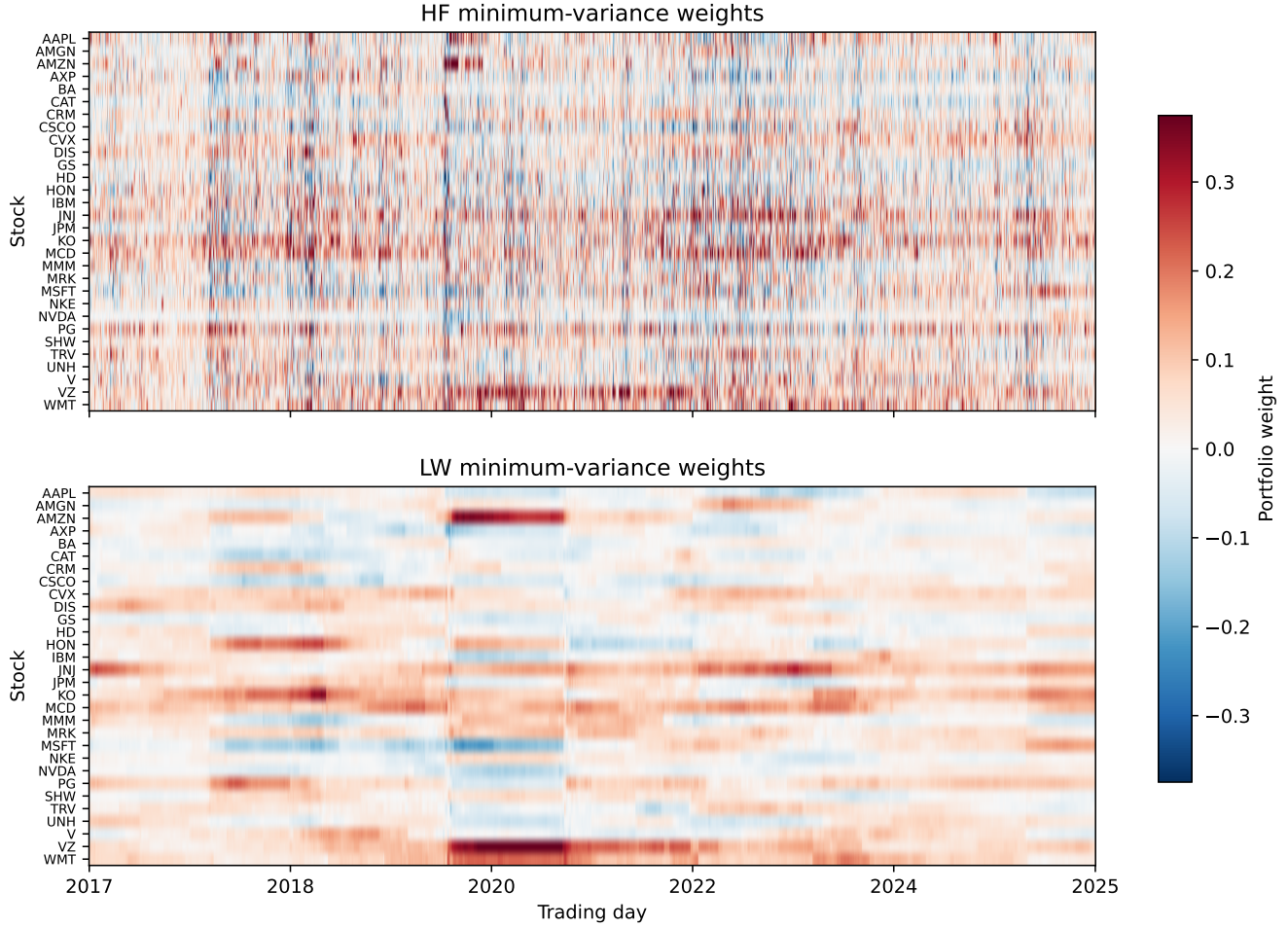


Figure 4: Minimum-variance portfolio weights for the high-frequency and Ledoit-Wolf covariance estimators. Rows are stocks, columns are rebalancing dates. Red indicates long positions, blue indicates short positions, and white indicates weights close to zero. Both panels share the same colour scale, capped at the 99th percentile of absolute weights so that typical variation is visible.

As shown in Figure 4, the high-frequency estimator produces more variable portfolio weights across stocks and rebalancing dates, while the Ledoit-Wolf estimator gives smoother and more persistent allocations. This reflects that the high-frequency estimator reacts directly to the previous day’s realized covariance matrix, whereas the Ledoit-Wolf estimator is based on a rolling window and shrinkage.

Table 1: Ten most extreme minimum-variance portfolio positions based on the high-frequency covariance estimator.

Trading day	Strategy	Stock	Weight (%)	Absolute weight (%)
2021-10-05	HF	VZ	148.1	148.1

Table 1: Ten most extreme minimum-variance portfolio positions based on the high-frequency covariance estimator.

Trading day	Strategy	Stock	Weight (%)	Absolute weight (%)
2020-03-30	HF	AMZN	133.97	133.97
2020-03-09	HF	AMZN	119.76	119.76
2020-03-19	HF	AAPL	119.02	119.02
2021-10-01	HF	VZ	118.94	118.94
2020-03-04	HF	HD	-116.91	116.91
2020-03-27	HF	AMZN	110.87	110.87
2020-04-13	HF	AMZN	110.12	110.12
2023-01-05	HF	JNJ	100.63	100.63
2021-10-07	HF	VZ	100.23	100.23

Table 2: Ten most extreme minimum-variance portfolio positions based on the Ledoit-Wolf covariance estimator.

Trading day	Strategy	Stock	Weight (%)	Absolute weight (%)
2021-02-17	LW	VZ	42.97	42.97
2021-02-16	LW	VZ	42.91	42.91
2021-02-11	LW	VZ	42.83	42.83
2021-02-12	LW	VZ	42.76	42.76
2021-02-10	LW	VZ	42.4	42.4
2021-02-09	LW	VZ	42.38	42.38
2020-12-30	LW	VZ	42.29	42.29
2021-01-05	LW	VZ	42.28	42.28
2020-12-29	LW	VZ	42.26	42.26
2021-01-06	LW	VZ	42.25	42.25

Table 1 and Table 2 report the ten largest absolute portfolio weights for each estimator. The high-frequency strategy generates the most extreme positions, with the largest absolute weight equal to `{python} top_extreme_abs_weight%` for `{python} top_extreme_stock` on `{python} top_extreme_day`. In contrast, the Ledoit-Wolf weights are more moderate, illustrating how shrinkage leads to more stable portfolio allocations.

The final part of the analysis evaluates how much the portfolio weights change between consecutive rebalancing dates. We measure rebalancing activity by daily turnover:

$$\frac{1}{T} \sum_t \sum_i |\omega_{i,t+1} - \omega_{i,t}|.$$

Turnover therefore measures how much trading is required to adjust the portfolio from one day to the next. A higher turnover indicates less stable portfolio weights and greater trading activity, whereas a lower turnover indicates more persistent portfolio allocations.

Table 3: Turnover of the high-frequency and Ledoit-Wolf minimum-variance portfolios.

Strategy	Average turnover (%)	Median turnover (%)	Max turnover (%)	Rebalances
HF	349.65	328.44	934.56	2222
LW	7.12	4.99	98.45	2222

As shown in Table 3, the high-frequency strategy has much higher average turnover than the Ledoit-Wolf strategy, with `np.float64(349.65)%` compared with `np.float64(7.12)%`. This reflects that the high-frequency covariance estimator reacts more strongly to new intraday information, while the Ledoit-Wolf estimator produces smoother covariance forecasts. The Ledoit-Wolf strategy would therefore require less rebalancing and likely incur lower transaction costs in practice.

References

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- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169.
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